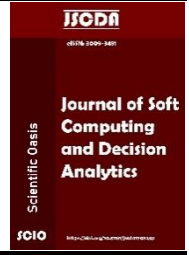




SCIENTIFIC OASIS

Journal of Soft Computing and
Decision AnalyticsJournal homepage: www.jscda-journal.org
ISSN: 3009-3481

A Comparative Study on Customer Churn Analysis Using Machine Learning and Data Enrichment Techniques

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ARTICLE INFO

Article history:

Received 1 November 2023

Received in revised form 19 April 2024

Accepted 1 June 2024

Available online 13 June 2024

Keywords:

Churn Analysis; Data Mining; Customer Relationship Management; Machine Learning Algorithms; RFM Analysis.

ABSTRACT

With the increasing amount of online shopping, companies can collect more customer data. Companies use this data to get to know their customers better and provide customized services. Churn analysis is one of the most essential analyses derived from the vast amount of data collected, which provides information about when a customer will stop shopping with the company. In this study, we perform a churn analysis using machine learning (ML) algorithms to analyse the customer behavior data of a fashion retail company. To perform churn analysis, we performed a four-stage methodology. First, we carried out data preparation and visualization studies, and then we created models using various ML algorithms. After examining the baseline data, we added the RFM (Recency, Frequency, Monetary) score to the data with the data enrichment technique and performed the analysis again. We used the Synthetic Minority Oversampling Technique (SMOTE) to eliminate the data irregularity and performed parameter optimization on the algorithms in SMOTE data. We compared the accuracy and F1 score values obtained after this four-stage process and examined the effect of the algorithms. In the last stage, we divided whole data into clusters using the k-means technique and applied ML algorithms to clustered data. Then, we compared all these results and examined the effect of segmentation on the results. The analysis shows that the extreme gradient boosting algorithm provides better accuracy and F1 score values. Using these results, the company can identify customers likely to churn and begin funding Customer Relationship Management (CRM) efforts. Additionally, experts can determine the company's development directions by organizing campaigns for these customers and analysing their reasons for churn in more detail.

1. Introduction

With the development of technology and the widespread use of the Internet, online shopping has become accessible to a broader audience. Online shopping offers the customer a much more comprehensive range of products than in stores and offers an effortless shopping opportunity without going to the store. Companies have the opportunity to offer their products to a broad

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E-mail address: eeisik@yildiz.edu.tr<https://doi.org/10.31181/jscda21202441>© The Author(s) 2024 | [Creative Commons Attribution 4.0 International License](https://creativecommons.org/licenses/by/4.0/)

audience. Another advantage for companies is that customer behavior can be analyzed by collecting more customer data.

This situation has been going on for a long time and is critical to many developments and renewals. For example, in the trade sector, analyzing customer behavior is very important. Customers are the people at the point where the companies' sales are concluded; understanding them is a factor that will directly affect their sales. It provides many advantages for companies to develop or take action according to the results of these analyses.

Customer behavior can be analyzed in many ways. Analyses are made in areas such as the frequency and spending of customers, the value they will provide to the company in the future, customer acquisition, and customer churn. All these review areas allow us to see customers from a different perspective. We should examine the client for whatever area we want to develop.

Due to the shopping habits of today's world, it is easy to gain a customer, but it is also very easy to lose it to another company. Since it is easier and more advantageous for a company to retain existing customers than to gain new customers, studies in this field are given importance.

Churn analysis predicts customers who show churn behavior and contributes to the creation of actions, campaigns, and discounts that will prevent customers from losing their customers. In this way, preventive actions aim to reduce the possibility of customer loss. When performing churn analysis, a rule set is created using values such as customers' purchasing frequency and the amount they spend. After the customers are identified, end-of-period calculations are made, and customers likely to churn are determined.

At the same time, with churn analysis, the impact of products on customers, their success, and customer continuity can be analyzed. Companies' interactions with their customers are not preferred to remain constant. It should be aimed at keeping the customer churn rates of growing companies very low and ensuring new customer acquisition. Thanks to churn analysis, future customer behavior is predicted by analyzing the current situation. For this purpose, campaign plans and practices are implemented. Using churn analysis, companies focus on the customer, who ensures continuity, provides accurate analysis of customer churn, and encourages customers to listen and analyze.

The retail industry is an area that is experiencing a vibrant process. This sector, where customers have a significant impact, also has high customer circulation. So much so that retaining existing customers is an issue that the entire industry considers and takes action on. Churn analysis can be very useful for companies in understanding customer behavior in this dynamic sector. Although there are many issues where churn analysis can benefit companies, its applications in the fashion retail industry are quite limited.

In this study, customer churn analysis was carried out using various algorithms and techniques on customer data. There are two phases in our research. In the first phase, we analyzed the data, and the accuracy values of the data were calculated using four algorithms without enriching the data. This analysis was carried out by enriching the data, oversampling, and parameter optimization processes. The algorithm/state that gives the ideal result is determined by comparing the results. In the second phase, the data were divided into clusters, and clusters were examined using related algorithms. As a result of the models, the customers that will churn are determined. Thus, it is possible to make the necessary strategic plans to turn these customers back into loyal customers.

In addition to its industrial contributions, this study will provide helpful information for studies on similar data sets. As a result of the study, it will be answered which of the ones included in the comparison is the most appropriate Machine Learning (ML) model to be used to predict future customer churn for the data set used and the parameters to compare these algorithms. Additionally, the contribution of data enrichment to the results and the effects of dividing the data into groups before the application can be examined.

This paper is organized as follows. In Section 2, we give a literature review of CRM, churn prediction, and ML algorithms that are commonly used for customer churn analysis. In Section 3, we introduce our methodology and explain our dataset. Section 4 presents our experimental results, and finally, Section 5 gives the overall results and managerial insights for managers.

2. Literature Review

Churn analysis, algorithms, and application examples are available in different sectors. In this section, literature research contributes to understanding the subject and presents previous applications. Additionally, the techniques used in the applications are also examined.

Churn analysis is critical in managing customer relations. Thanks to this analysis, customers who are dissatisfied with the company or have lost interest in the company can be identified, and strategies for these customers can be determined [1]. Gil-Gomez et al., [2] mentioned the definition of Customer Relationship Management (CRM), its application methods, contributions to innovation, and CRM strategies. Matuszelański and Kopczewska's study [3] presents the effects and importance of gaining customers, loyalty, and churn prediction on customer relationship management. After data analysis, CRM and the results that can be done on this subject were examined.

Kaynar et al., [4] provided data collection, CRM applications, and model setup information. The churn analysis study established the model using several methods: Support Vector Machine (SVM), Naive Bayes, and Artificial Neural Network (ANN). Hamdy and Kandel [5] performed a comparative churn analysis on customer data of a telecommunication company. The best results from this article were obtained after applying pre-processing techniques. Thus, the problems of missing values, high cardinality, and high dimensionality were reduced. Celik and Osmanoğlu [6] compared techniques used for churn prediction. The authors also included many definitions in their article, such as churn, its steps, techniques, and ML algorithms.

Cooper [7] gave details about employee churn analysis. Four different classification models were established on the employee cluster. Naive Bayes, SVM, Decision Tree, and ANN techniques were applied. Then, these methods were evaluated by using statistical analysis (ANOVA). In Wu et al.'s study [8], churn analysis is applied with different methods for customer prediction. The authors divided the customers into clusters using the k-means algorithm and established the model using the algorithm.

Dingli et al., [9] applied and compared the deep learning algorithms, Convolution Neural Network and Restricted Boltzmann Machine, on the local retail industry data. The authors compared the results according to accuracy values. Asthana [10] processed data using a boosting method that supports ML. The author used Monte Carlo simulation and examined the most efficient parameter combinations. A comparison was made with the previous data, and it was found that the amplified data gave better results for churn.

An example of a study that consumes transactional data churn prediction was given by Aleksandrova [11]. This study performed churn analysis by applying ML techniques to the transactional data with RFM analysis. In Ahmad Naz et al.'s study [12], modeling techniques such as Logistic Regression (LR), Neural Network (NN), Decision Tree, SVM, and Naive Bayes were applied to common customer data. Then, the best results in the desired category are listed. If a better true churn rate and false churn rate are expected, Decision Tree and SVM should be used, and if churn probability is concerned, LR should be used according to the results.

Sucki [13] established different models on different datasets. Random-Forest (RF) and AdaBoost give the best results among these models. While examining the algorithms, the author used comparison matrices, ROC-curve, F-score, accuracy, and AUC scores. In Wadikar's study [14], churn was estimated using different algorithms: LR, RF, SVM, and NN. After applying these algorithms to

the unbalanced dataset, they were also used to balance the data with the SMOTE technique, and the results were compared. In another study, Makruf et al., [15] applied four different algorithms (ANN, SVM, K-Nearest Neighbors, Naive Bayes, and Decision Tree) to the customer data. After the application, the accuracy and F-scores of the algorithms were calculated and compared.

Patel et al., [16] defined retail churn as the total number of store openings and closings associated with a firm during a quarter and used dynamic decision-making to explain why retail store churn. Geiler et al., [17] compared eight supervised ML methods on thirteen public churn data sets. The authors identified the most appropriate methods in an attrition context and emphasized that their strategy can give information to marketing or human resources services on the behaviors of customers. Baghla and Gupta [18] compared four ML techniques for predicting customer churn in e-commerce. The authors implemented those techniques on the Brazilian e-commerce dataset and measured their performance using six performance metrics.

Prabadevi et al., [19] studied to find the optimum ML strategy for early client churn prediction. The authors tested stochastic gradient booster, random forest, logistics regression, and k-nearest neighbors methods and discussed the best among the four from different perspectives. Shobana et al., [20] proposed a process for forecasting e-commerce customer attrition based on SVM and presented a hybrid recommendation strategy for targeted customers. Haddadi et al., [21] presented a comprehensive analysis of customer churn prediction focusing on highly imbalanced datasets. The authors employed the comparative analysis regarding fourteen distinct classification methods considering resampling strategies, like SMOTE. Singh et al., [22] examined bank data to forecast which customers will most likely churn. To do this, the authors used various ML algorithms to compare different evaluation metrics. Although various studies have been conducted on comparing different ML algorithms for churn analysis, studies investigating the effect of data enrichment and prior clustering on the performance of these algorithms are limited. These analyses have not been applied before in the fashion retail industry. Our study focuses on these gaps in the literature.

3. Material and Method

Before the implementation phase, it is necessary to understand the study's subject and the examination method. The method used in the study and the data examined is explained in the following subsections.

3.1 Theoretical Method

In this study, we made an analysis using four algorithms: RF, XGBoost, LR, and Naive Bayes. The main reason we chose these algorithms is that they are easy to apply and fast in general. In addition, they have frequently used algorithms, so we can easily access resources and previous examples. Before applying algorithms, it is crucial to understand the basics of algorithms. In this section, we explained the definition, the stages of implementation, and the underlying ideas of the algorithms we applied.

RF is an algorithm for prediction, regression, and classification using many decision trees. With this algorithm, an improved version of decision trees, the overfitting problems of the decision tree method are resolved. A random forest can be created easily due to its simplicity. Therefore, its use is quite common. The previous studies also show a good result in terms of performance. It stands out among the algorithms that will perform better due to its speed. They can use many data features, categorical, numeric, and binary [23].

First, it is necessary to understand the XGBoost algorithm to understand decision trees and gradient boosting. Decision trees are algorithms that create a set of rules to predict the outcome of an event. For example, by building an illness model, we can predict whether a person will recover

from it or recover time. On the other hand, the gradient boosting method enables the creation of a new model using several models. After the first model is created, the second model is created using the first model. For the third model, the first and second models are weighted according to their performance, and in this way, the model continues to be built. Thus, an efficient model is tried to be created. The XGBoost algorithm is created by combining these decision trees and gradient-boosting methods. They are designed for optimum performance and speed. XGBoost can be expressed as a supervised Decision Tree learning algorithm that applies boosting to produce reasonably accurate models. XGBoost is highly predictive and runs ten times faster than other decision tree-based algorithms [24].

Regression analysis is a method used to estimate the relationship between dependent and independent variables. It is an easy-to-apply, broad, and frequently used analysis. There are many types of regression. The most widely used of these are linear and logistic regression. In linear regression, it is considered that there is a linear relationship between the input and output values, and the model is built accordingly. In logistic regression, the outcome of the dependent variables is binary. One of the two values should appear as a result of the model, and the model is set up accordingly. The main purpose of logistic regression is to establish the model that best fits the data. The number of target clusters is certain in logistic regression, and binary classification is applied. Logistic regression can describe the relationship between the binary outcome variable and the independent variables consisting of continuous and discrete variables [25].

The Naive Bayes algorithm is based on the Bayes theorem, one of the essential statistics topics. This theorem shows the relationship between conditional and marginal probabilities within the probability distribution for a random variable. An algorithm to be applied to statistical classification problems is obtained in the model developed [26].

After the data was analyzed using data enrichment, SMOTE method, parameter optimization, and clustering, these four algorithms were used. Each output's precision, recall, F1 score, and accuracy values were checked to evaluate the outputs of the models used in the study. Each of these outputs can gain meaning according to the purpose of the study and the data structure. We obtain these values from the following confusion matrix.

Table 1

Confusion Matrix

	Predicted	
	True Positives	False Negatives
Actual	False Positives	True Negatives

True positive and true negative values give the number of data that our model predicted correctly. As given in Table 1, a false positive is a Type I error, and our model predicted that these customers would not churn, but these customers are actually churn. If it's a false negative, it's a Type II error, and our model predicted customers who wouldn't churn would be churn. However, precision shows how many of the values we estimated as Positive are actually Positive.

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

On the other hand, recall is a metric that shows how much of the operations we need to estimate as Positive; we estimate as Positive.

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

The F1 Score value shows us the harmonic mean of Precision and Recall values.

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

The ratio of the areas calculates the accuracy value we predict correctly in the model, that is, the total true values to the complete data set;

$$\frac{TP + TN}{TP + FP + TN + FN} \quad (4)$$

3.2 Results

The real-world dataset belongs to a global fashion retail company. Colin's is a retail trade company serving the women's/men's clothing industry. The company is one of the well-known brands in the field of Türkiye. This data, including the pandemic period, is the CRM data of Colin's company. In Figure 1, the distribution of customers by country is given. We used only CRM data of Turkish customers in the study.

```
ulke = pd.crosstab(index=df5['ULKE'], columns='Proportion')
ulke/ulke.sum()
```

col_0	Proportion
ULKE	
BELARUS	0.031206
ROMANYA	0.005284
RUSYA	0.392146
TANIMSIZ	0.000174
TURKIYE	0.444457
UKRAYNA	0.126733

Fig. 1. Dataset Country Distribution

Since the shopping habits of the citizens of each country can change and different geographies consist of different cultures, data from other countries were removed from the dataset at the beginning of the study.

```
df_11 = df1[(df1['ULKE'] == "TURKIYE") & (df1['SON_AV_TARIH'] < 365 )]
df_22 = df2[(df2['ULKE'] == "TURKIYE") & (df2['SON_AV_TARIH'] < 365 )]
df_33 = df3[(df3['ULKE'] == "TURKIYE") & (df3['SON_AV_TARIH'] < 365 )]
df_44 = df4[(df4['ULKE'] == "TURKIYE") & (df4['SON_AV_TARIH'] < 365 )]
df_55 = df5[(df5['ULKE'] == "TURKIYE") & (df5['SON_AV_TARIH'] < 365 )]
df_66 = df6[(df6['ULKE'] == "TURKIYE") & (df6['SON_AV_TARIH'] < 365 )]
df_77 = df7[(df7['ULKE'] == "TURKIYE") & (df7['SON_AV_TARIH'] < 365 )]
df_88 = df8[(df8['ULKE'] == "TURKIYE") & (df8['SON_AV_TARIH'] < 365 )]
```

Fig. 2. Data Elimination

The company keeps very detailed CRM data, as seen in Figure 2. The dataset consists of 67 columns in total; one of these columns is the unique ID, and the remaining columns constitute the variables. The data has few categorical demographic variables; all other data are numerical. The customers' shopping habits were examined according to their time breakdown, the types of products they bought, their shopping frequency, and the channels they shopped to manage customer relations. In addition, there is some information about the customer, such as gender and age. The target column for churn in this study is the CHURN_SONRAKI_1YIL variable.

4. Experimental Results

After all the arrangements on the data were completed, the dataset was inserted into all four estimation models used in the study. As a first phase, outputs were taken from the base models. Then, the outputs obtained from each model were tried to be improved using different techniques; the process diagram of the whole methodology is given in Figure 3.

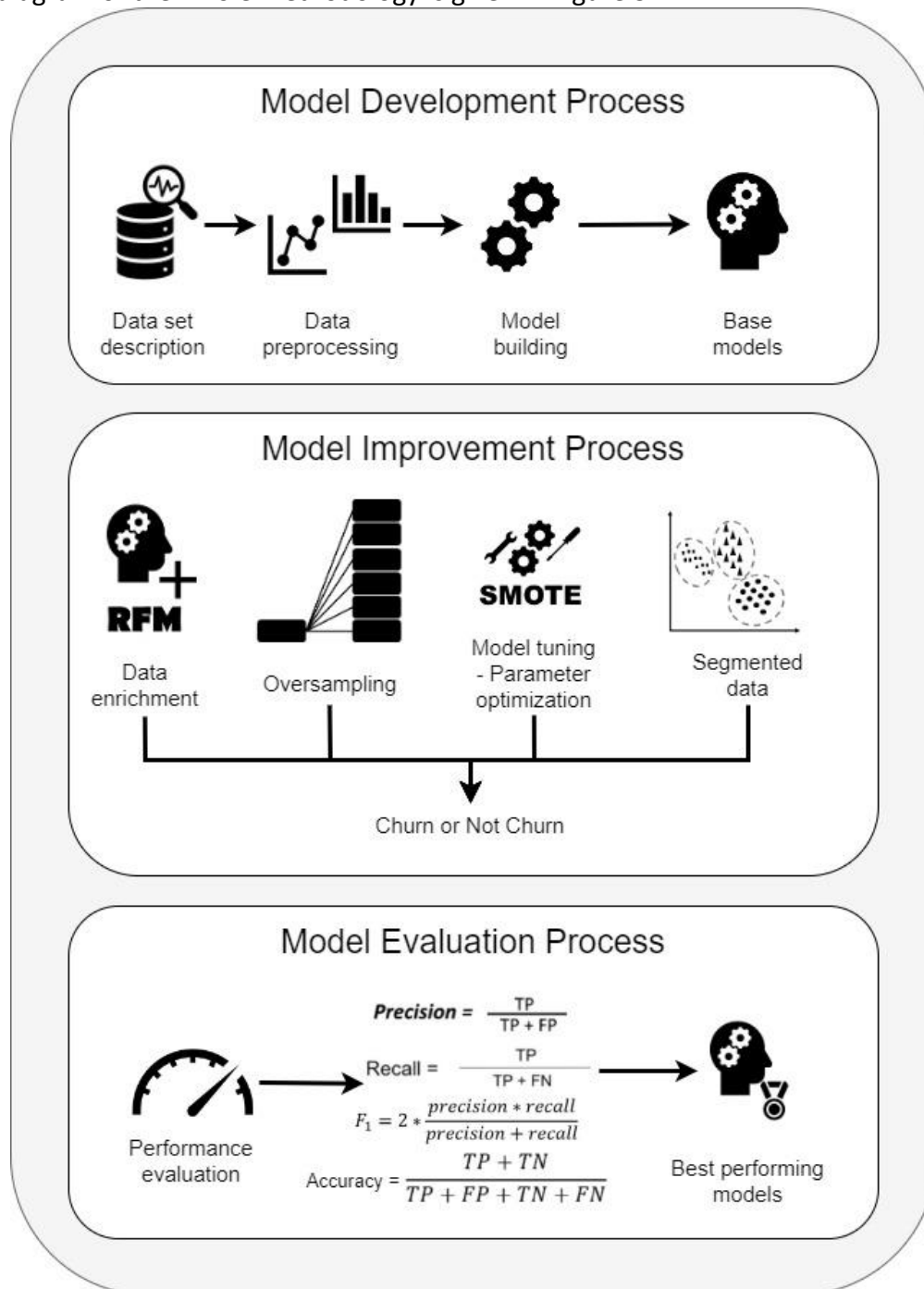


Fig. 3. The process of the whole methodology

4.1. Results and Analysis for Experiment with Whole Data

In the first phase, the data were analyzed without division. XGBoost, RF, LR, and Naive Bayes algorithms were applied to the base data, the data enriched with the RFM score, the data with the irregularity removed by the SMOTE method, and the parameter-optimized data, respectively. In models where the whole dataset is taken, it would be more accurate to look at the F1 score instead

of accuracy in the base model and enriched model. Because in these models, the distribution of the target column is uneven. Table 2 shows the comparative analysis of ML models based on the F1 score.

Table 2
Comparative Analysis of ML Models Based on F1 Score

F1 Score		XGBoost	Random Forest	Logistic Regression	Naïve Bayes
Base Model	Not Churn	0.41	0.44	0.41	0.5
	Churn	0.82	0.81	0.82	0.77
Enriched Data	Not Churn	0.41	0.44	0.41	0.52
	Churn	0.83	0.8	0.82	0.75

According to Table 3, the XGBoost and Random Forest algorithms successfully identified our customers who would churn in the base models. At the same time, the Naive Bayes method performed better in identifying the customers who would not churn with the same dataset.

Table 3
Comparative Analysis of ML Models Based on Accuracy Score

Accuracy	XGBoost	Random Forest	Logistic Regression	Naïve Bayes
Oversampling	0.72	0.70	0.69	0.66
Model Tuning	0.73	0.73	0.69	0.70

After producing synthetic data by oversampling, our data set reached a balanced point in terms of distribution in the target column. When the accuracy was evaluated after the oversampling and model tuning stages, according to Table 3, the model established with the XGBoost algorithm gave better results than the other models.

4.2. Results and Analysis for Experiment with Clustered Data

We first took only the RFM values to divide the data into clusters. The Elbow method determined the number of clusters, and data were distributed to these clusters with RFM values. According to the elbow method, the number of clusters was determined as four. Accordingly, the customers were divided into segments, represented by clusters we obtained. The created clusters are given in Figure 4 as 3D.

At this stage, all clusters were included in the models separately, and the models were found to have independent prediction accuracy. In addition, the models showed different performances for the same clusters.

Table 4 examines the number of accurate predictions for each model in total for all four clusters as a performance metric. At this point, while Naive Bayes provided the highest number of correct predictions in cluster2, it lagged behind the performance of other models in general. XGBoost and Logistic Regression models showed the best performance regarding the total number of accurate predictions. Segmentation-based churn analysis is one of Colin's ultimate goals, as is the dataset we obtained. The implementations we have done in this study, the techniques used, and the outputs will contribute to the company's determination of the customers likely to churn with high accuracy.

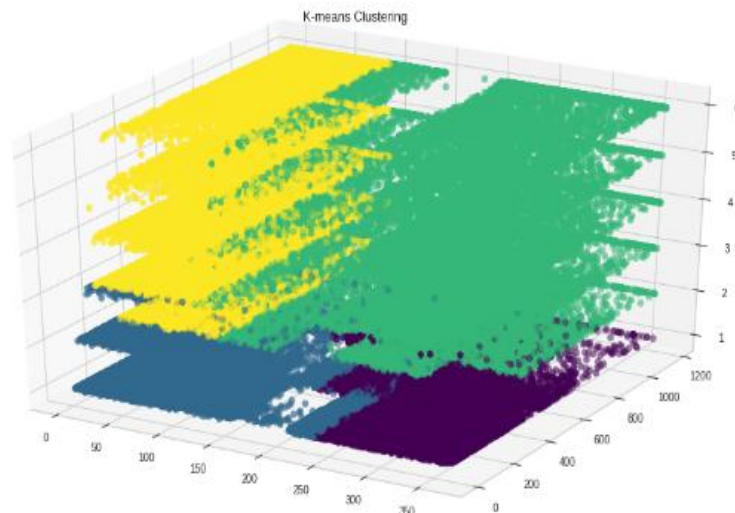


Fig. 4. Clusters' 3D View

Table 4
Comparative Analysis of ML Models for Segmented Data

True Predictions	XGBoost	Random Forest	Logistic Regression	Naïve Bayes
Cluster0	4770	4655	4828	4368
Cluster1	5171	4930	5171	4432
Cluster2	3922	3922	3922	3982
Cluster3	3303	3251	3251	2150
Total	17166	16797	17171	14932

5. Conclusion

This study aimed to provide the best estimation algorithm and data state to estimate the customer churn rate. For this purpose, the customer behavior data of a fashion retail company were analyzed. When the whole data was analyzed firstly, i) four different algorithms were applied to the base data, ii) the data irregularity was tried to be eliminated by using the SMOTE technique, iii) the data was enriched by adding the RFM score value to the base data, iv) four algorithms were applied by making parameter optimizations in the base data. After examining the data as a whole, in the second phase, the data was divided into clusters using the k-means algorithm according to customer characteristics, and algorithms were applied to these clusters. Algorithms' performances were measured using two performance indicators: F1 score and accuracy. Comparisons were made with those performance indicators for all algorithm implementations to determine the best analysis technique and algorithm.

The models were run five times while implementing the four algorithms for optimal results. While the results of the obtained algorithms were quite close to each other, the Extreme Gradient Boosting algorithm showed better results than the others. The ML models established in the study accurately predict the probability of the company's customers from which the data is taken to be churn, approximately 73%. After the company determines the customers that possibly churn, they can start providing resources for the CRM studies we explained in the first parts of the study. Making campaigns for these customers, analyzing the reasons for leaving these customers, and determining the development directions of the company form the basis of the use of the results obtained after the churn analysis.

Although our study produced very good results in the applied field, different methods can be applied in future studies to obtain better results. Methods that provide good results in many areas, especially deep learning, can be tried in appropriate operating environments. Additionally, in future studies, a study focused on performing churn analysis and creating particular strategies for that group can be made after customer data is divided into clusters with RFM data, as in the examination method in the last phase. In fact, instead of dividing the segments directly into 5-6 groups, it is possible to create a common strategies framework by naming them with classical RFM groups and blending the strategies designed after RFM with the strategy to win the customers who will churn.

Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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